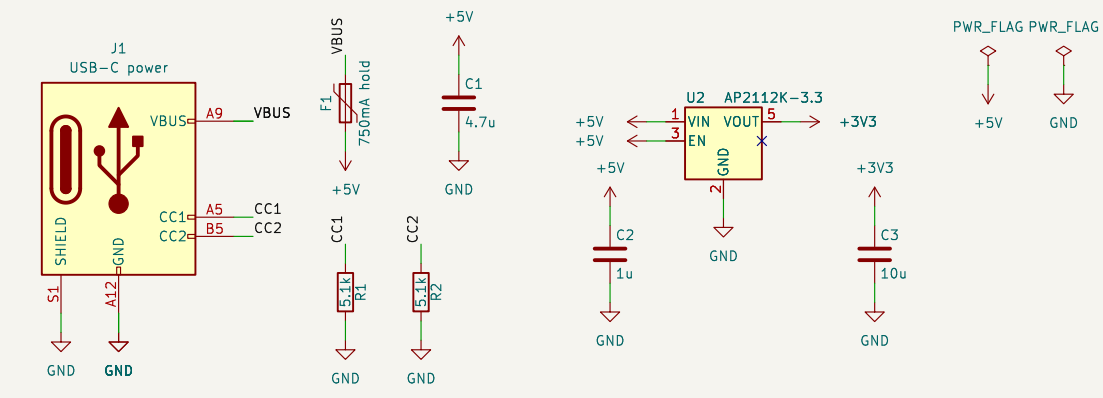
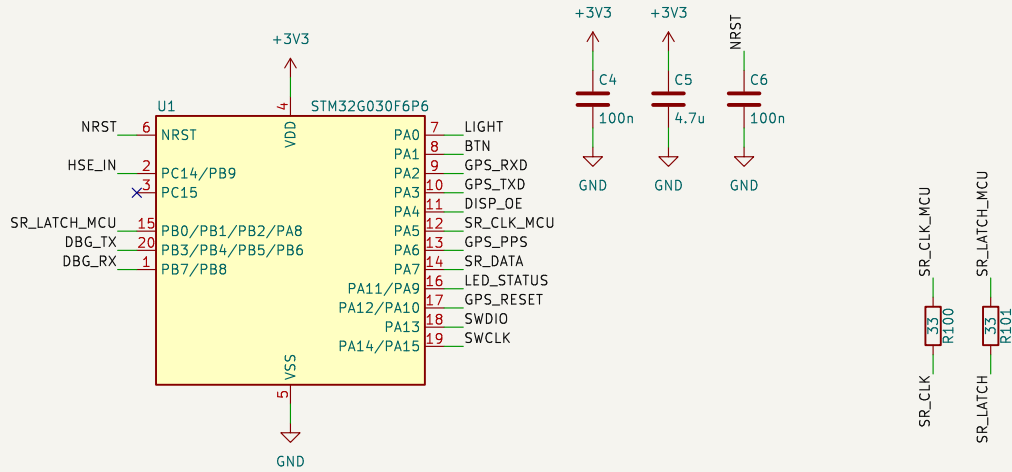


POWER: USB-C 5 V in, power only



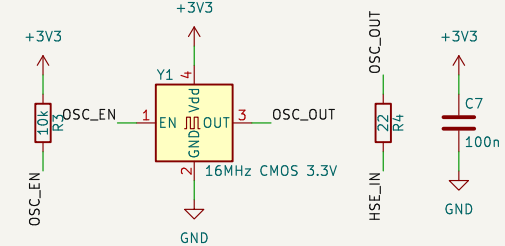
MCU: STM32G030F6P6 (TSSOP-20)

HSE in BYPASS mode on PC14: this package has no OSC_OUT, so no crystal.



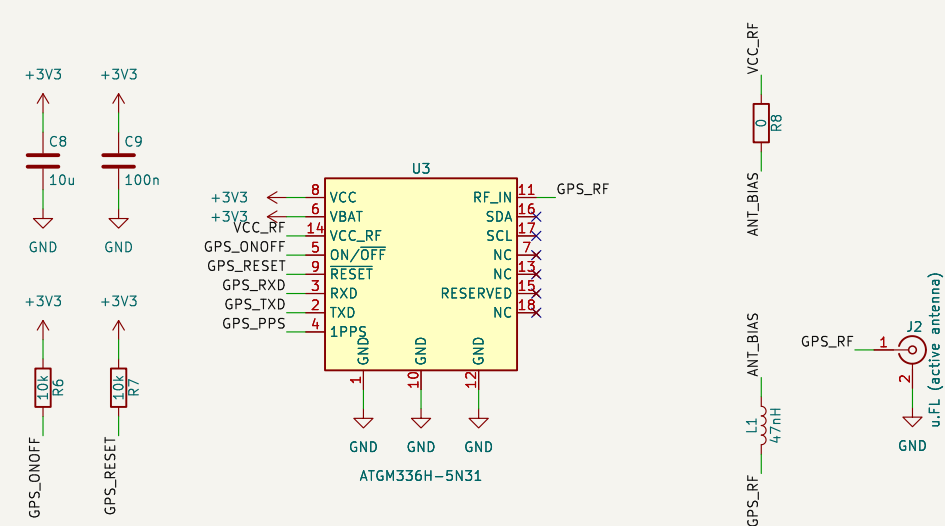
Clock: 16 MHz 3.3 V CMOS oscillator.

Same footprint fits a CMOS-output TCXO (check its pin 1 is EN or NC, not VC).

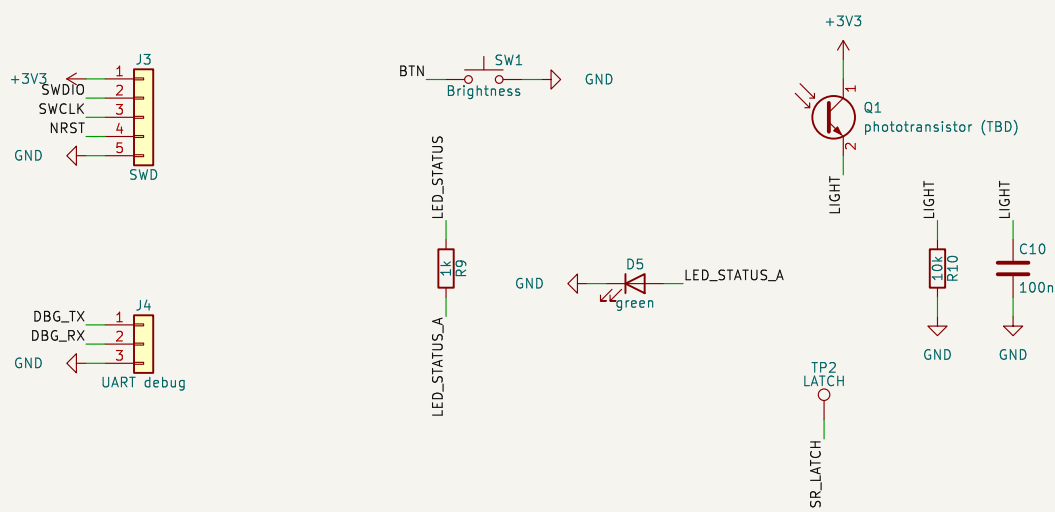


GPS: ATGM336H-5N31

Uses the u-blox MAX footprint (same 18-pad layout). Active antenna via u.FL.



I/O: SWD, debug UART, button, status LED, light sensor (optional)



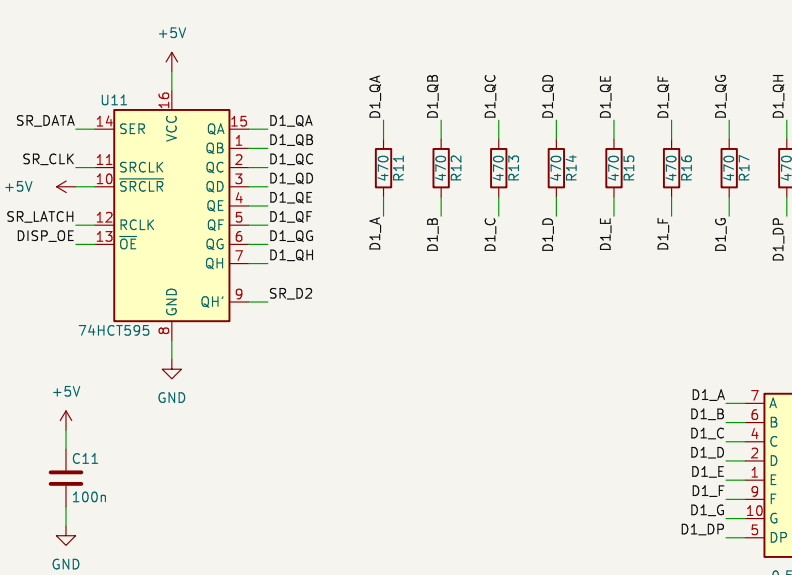
DISPLAY: HH:MM:SS.mmm, 9 digits, each statically driven by a 74HCT595 (5 V).

Chain: SR_DATA -> U11 -> U12 ... -> U19. All latch together on SR_LATCH. /OE = PWM brightness.

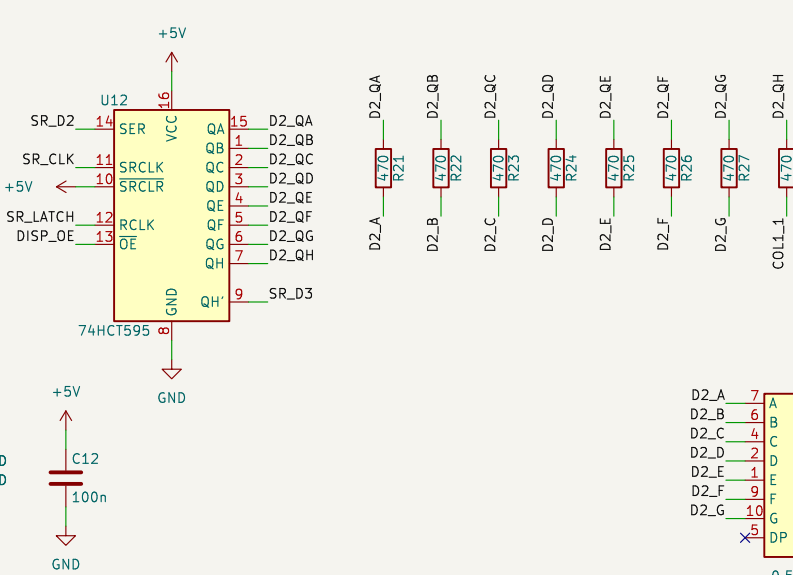
Segment resistors 470R: ~5.7-6.4 mA per LED. 8 LEDs per 595 (~46-51 mA); U12/U14 drive 9 (~51-58 mA). Limit 70 mA.

All 74 LEDs on: ~0.42-0.47 A from 5 V. Firmware must cap /OE duty for an all-segments self-test.

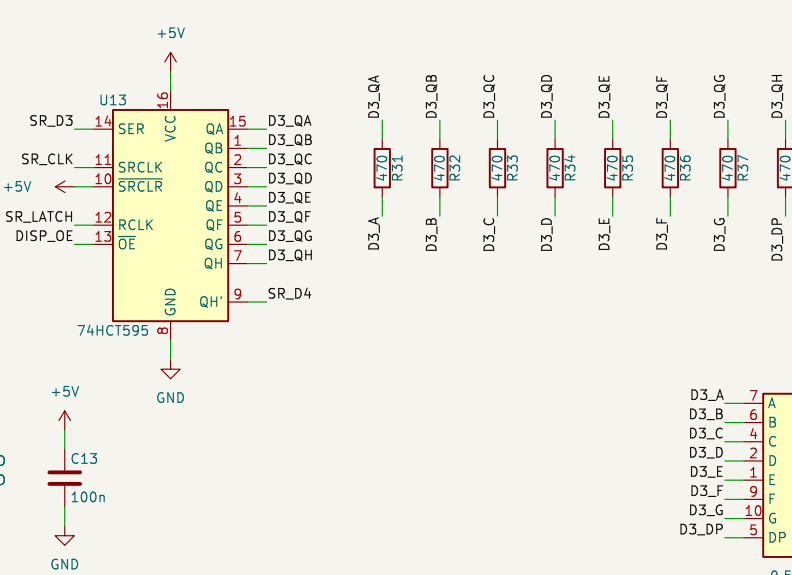
DS1: H tens



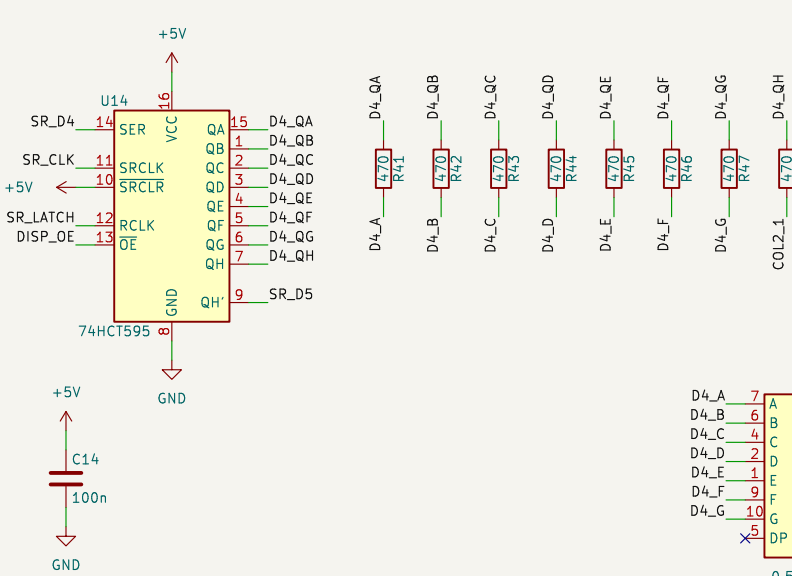
DS2: H units



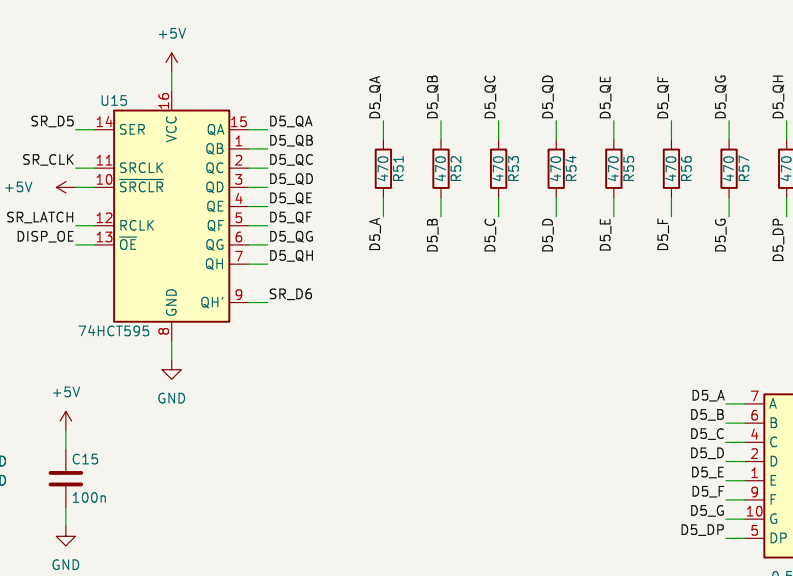
DS3: M tens



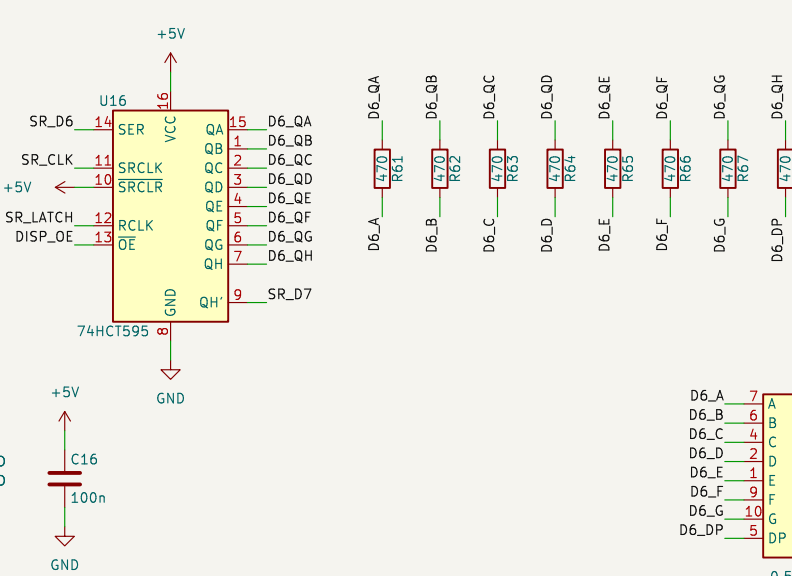
DS4: M units



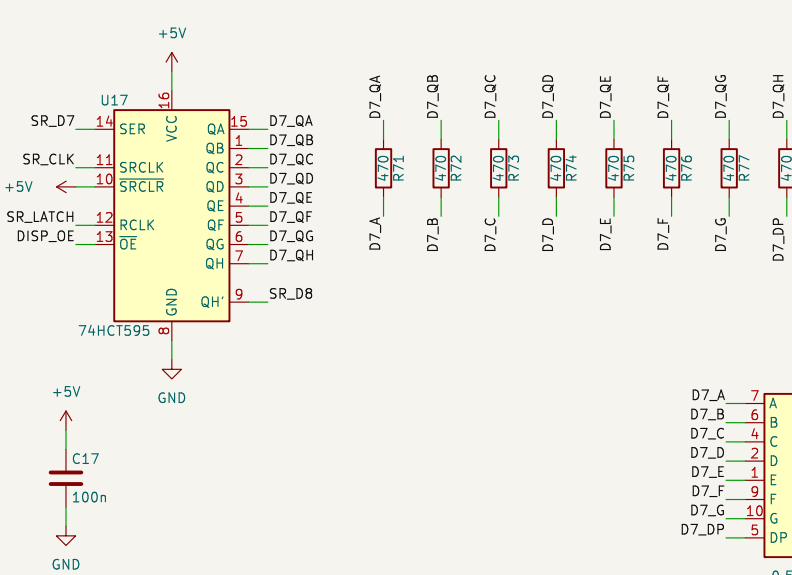
DS5: S tens



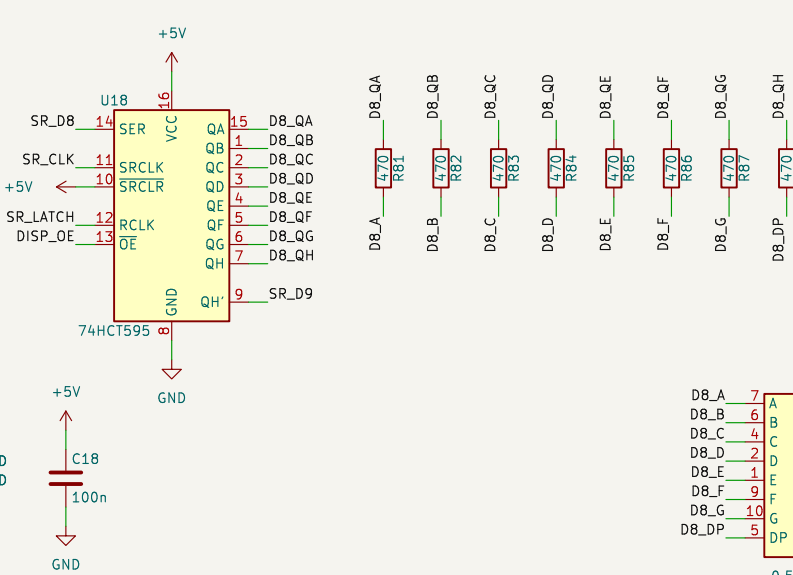
DS6: S units (DP = decimal point)



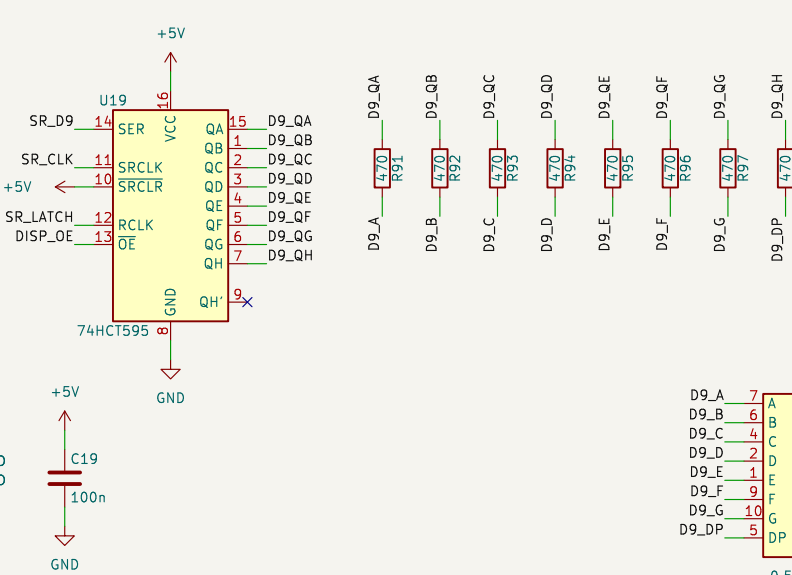
DS7: 1/10 s



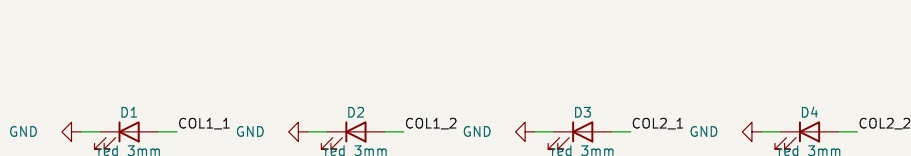
DS8: 1/100 s



DS9: 1/1000 s



Colons: between DS2|DS3 (COL1) and DS4|DS5 (COL2). Two 3 mm red LEDs each, driven from QH of U12 / U14.



Generated by hardware/tools/gen_schematic.py – edit the script, not this file

Sheet: /
File: gpsclock.kicad_sch

Title: GPS ms wall clock

Size: A2 Date: 2026-10-04

KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4

Rev: 0.1

Id: 1/1